LINKEDIN

- Company: LinkedIn
- Industry: Professional Networking & Technology
- Use Case: Customer Support Agentic System.
- Reference: LangChain/LangGraph Case Studies: LinkedIn Support

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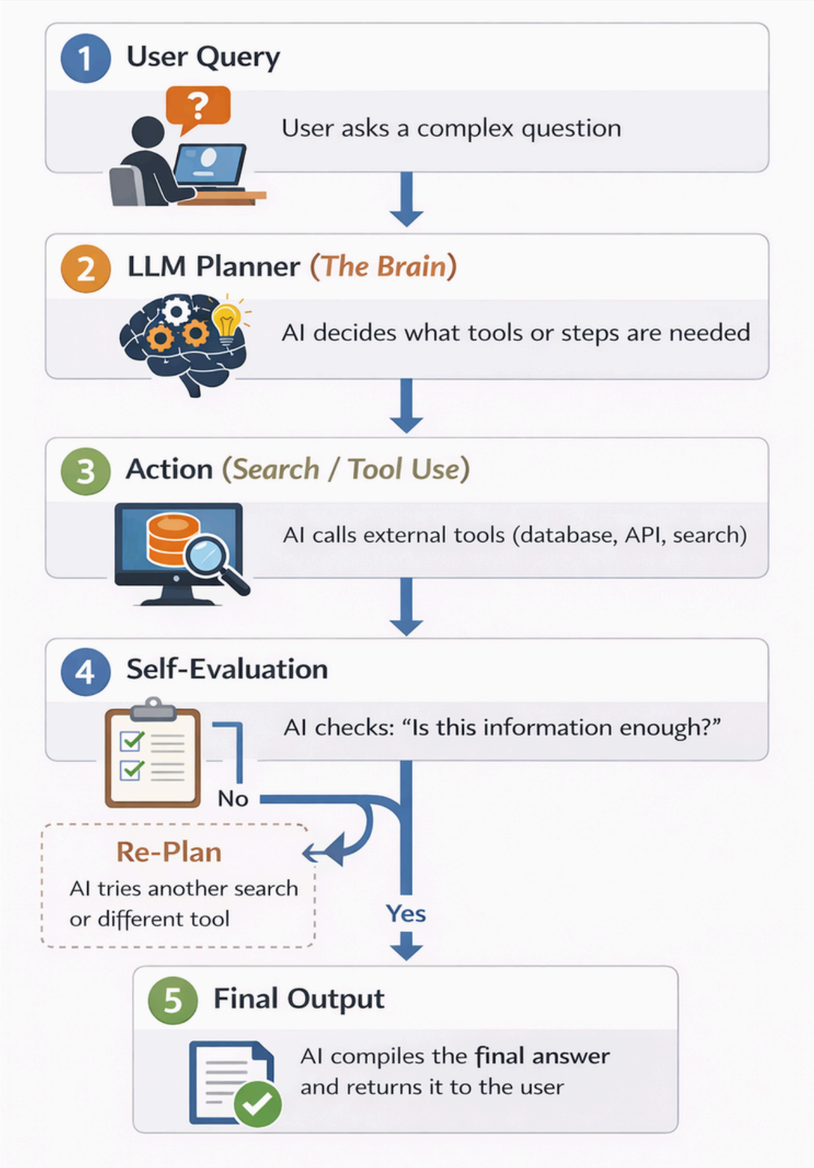
THE CHALLENGE & THE SOLUTION

- The **Problem**: LinkedIn handles a massive volume of support tickets. Traditional chatbots were too rigid, failing to solve complex technical issues that required multi-step reasoning or searching through deep documentation.
- The **Solution**: LinkedIn implemented an Agentic RAG (Retrieval-Augmented Generation) system using LangGraph.
- **Objective**: To move beyond simple "Question-Answer" bots to an "Agent" that can navigate documents, reason about the user's specific problem, and verify its own answers before responding.

AGENTIC VS. NON-LLM COMPONENTS

Agentic (LLM-Driven)	Non-LLM (Software Components)
Reasoning & Routing: The LLM decides *which* internal tool or document set to search based on the user's intent.	**Vector Database (Elasticsearch):** The software engine that stores and retrieves raw text chunks via mathematical similarity.
Self-Correction: Using LangGraph "cycles" to evaluate if the retrieved info is relevant. If not, the LLM decides to search again.	**Authentication & Permissions:** Standard code that ensures the user only accesses data they are authorized to see.
Tone & Synthesis: Formatting the final technical solution into a professional, human-friendly response.	**API Gateway:** The infrastructure that connects the chat interface to the backend servers.

THE AGENTIC WORKFLOW



THANK YOU!